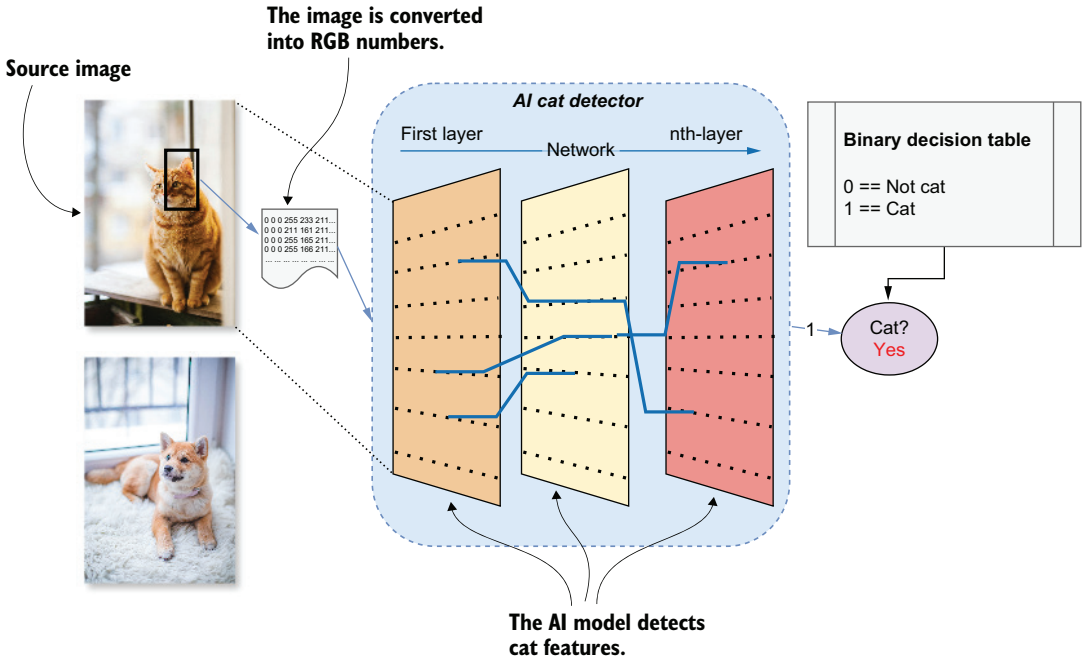


Figure 1.5 AI can be used to detect whether a picture of a cat is, in fact, a cat. It accepts an image as input and responds with *yes* or *no* (or 0 and 1).

Generative AI expands on machine learning's ability to detect patterns. Rather than simply recognizing or classifying data, generative models learn underlying patterns deeply enough to produce entirely new content that resembles the material they were trained on. They can mimic sentence structures, design styles, and musical patterns, generating results that previously required manual human effort.

The development of transformer architectures has propelled generative AI to its current capabilities. The field of AI in computer technology goes back a long time, but two recent developments in the models have revolutionized it: transformers and their core component, self-attention, both introduced in the landmark paper by Vaswani et al., "Attention Is All You Need" (https://arxiv.org/abs/1706.03762).

Transformers are a class of deep learning models that use self-attention mechanisms to process input data in parallel, making them highly effective in capturing long-range dependencies and contextual information within sequences. For example, in a sentence like "The lion, which had been chasing the antelope, finally caught it," transformers can understand that "it" refers to "the antelope" even though several words separate them.

Self-attention enables this understanding by allowing the model to evaluate the relative importance of each word in a sequence. Instead of viewing words in isolation, the model examines the full context to determine which words influence one another. As